

Artillery Load Test Report

Employee Document Vault — Member 4: Cost + CI/CD + Load Test

Item	Recorded value
Test endpoint	GET /files
Environment	AWS us-east-1, API Gateway prod
Target	https://cidr0fzgt5.execute-api.us-east-1.amazonaws.com/prod
Authentication	Cognito bearer token (stored locally; not included in this report)
Load tool	Artillery 2.0.33
Controlled baseline	10 maximum virtual users
Load phase	60 seconds
Artillery summary timestamp	16 Sep 2026, 00:29:57 (+05:30)

1. Objective

Validate the behavior of the production Employee Document Vault document-listing endpoint under a controlled baseline load, and capture end-to-end latency, HTTP success/error behavior, Lambda-side errors and throttling, concurrency, and X-Ray visibility.

2. Test Scope and Important Assumption

The supplied production-hardening sprint guide contains a missing/garbled value for the original concurrency requirement. Therefore, this report does not claim that 10 concurrent users was the original assignment requirement. A controlled baseline of 10 maximum virtual users was used so that the system could be exercised and measured without inventing the missing requirement.

The test exercised GET /files using a valid Cognito-authenticated request. A direct PowerShell validation returned HTTP 200 before the final Artillery run, confirming that authentication and the endpoint were functioning.

3. Artillery Test Configuration

```
config:
  target: "https://cidr0fzgt5.execute-api.us-east-1.amazonaws.com/prod"
  phases:
    - duration: 60
      arrivalRate: 10
      maxUsers: 10
      name: "Controlled 10-VU baseline"

  scenarios:
    - name: "List employee documents"
      flow:
        - loop:
          - get:
              url: "/files"
            headers:
              Authorization: "Bearer {{ $env.AUTH_TOKEN }}"
            - think: 1
          count: 60
```

4. Artillery Results

Metric	Result
HTTP 200 responses	600
HTTP errors	0
Total requests	600
Request rate	5 requests/sec
Virtual users created	10
Virtual users completed	10
Virtual users failed	0
Minimum response time	236 ms
Mean response time	440.2 ms
Median response time	424.2 ms
P95 response time	685.5 ms
P99 response time	1,224.4 ms
Maximum response time	1,396 ms

The Artillery process took approximately 1 minute 35 seconds overall because scenario iterations continued completing after the 60-second load phase. The configured load phase itself was 60 seconds. The 600 HTTP requests were all successful.

5. AWS CloudWatch Verification

Metric	Observed
Lambda invocations in selected window	605
Lambda errors	0
Lambda success rate	100%
Lambda throttles	0
Maximum concurrent executions	10
Average Lambda duration	215 ms
Maximum Lambda duration	398 ms

CloudWatch showed 605 invocations in the selected monitoring window, while Artillery generated 600 requests for the load test. The difference indicates that the CloudWatch window also contained a small amount of other application activity. The load-test request count is therefore reported from Artillery, not inferred from the broader CloudWatch window.

6. X-Ray Verification

The X-Ray Trace Map showed the expected request path: Client → API Gateway prod → listDocuments Lambda Context → listDocuments Lambda Function. This confirms that tracing was active and that the tested API path was visible through the observability layer.

7. Performance Assessment

At the controlled 10-VU baseline, the endpoint completed all 600 authenticated requests with HTTP 200 responses and no failed virtual users. No Lambda errors or throttles were observed. The mean end-to-end response time was 440.2 ms, with P95 of 685.5 ms and P99 of 1,224.4 ms. The higher P99 compared with the mean indicates some tail-latency variation, although there was no evidence of outright failure or throttling at this baseline.

8. Limitations

This was a controlled baseline rather than a capacity limit test. The original assignment's concurrency value was absent from the supplied guide, so 10 VUs was used only as a documented test assumption. The test therefore cannot establish the application's maximum supported concurrency. A higher-load test should be run once the intended concurrency requirement is known.

9. Recommendations

- Repeat the test at the actual assignment concurrency once that missing requirement is confirmed.
- Perform staged higher-load tests to identify the point where P95/P99 latency rises materially or errors/throttles appear.
- Continue monitoring Lambda duration, errors, throttles, API Gateway 4XX/5XX, and X-Ray traces during higher-load tests.
- Consider enabling API Gateway access logging and detailed metrics if additional request-level operational visibility is required.
- Treat the 10-VU results as the baseline against which future higher-load runs can be compared.

10. Evidence

A. Artillery final run summary

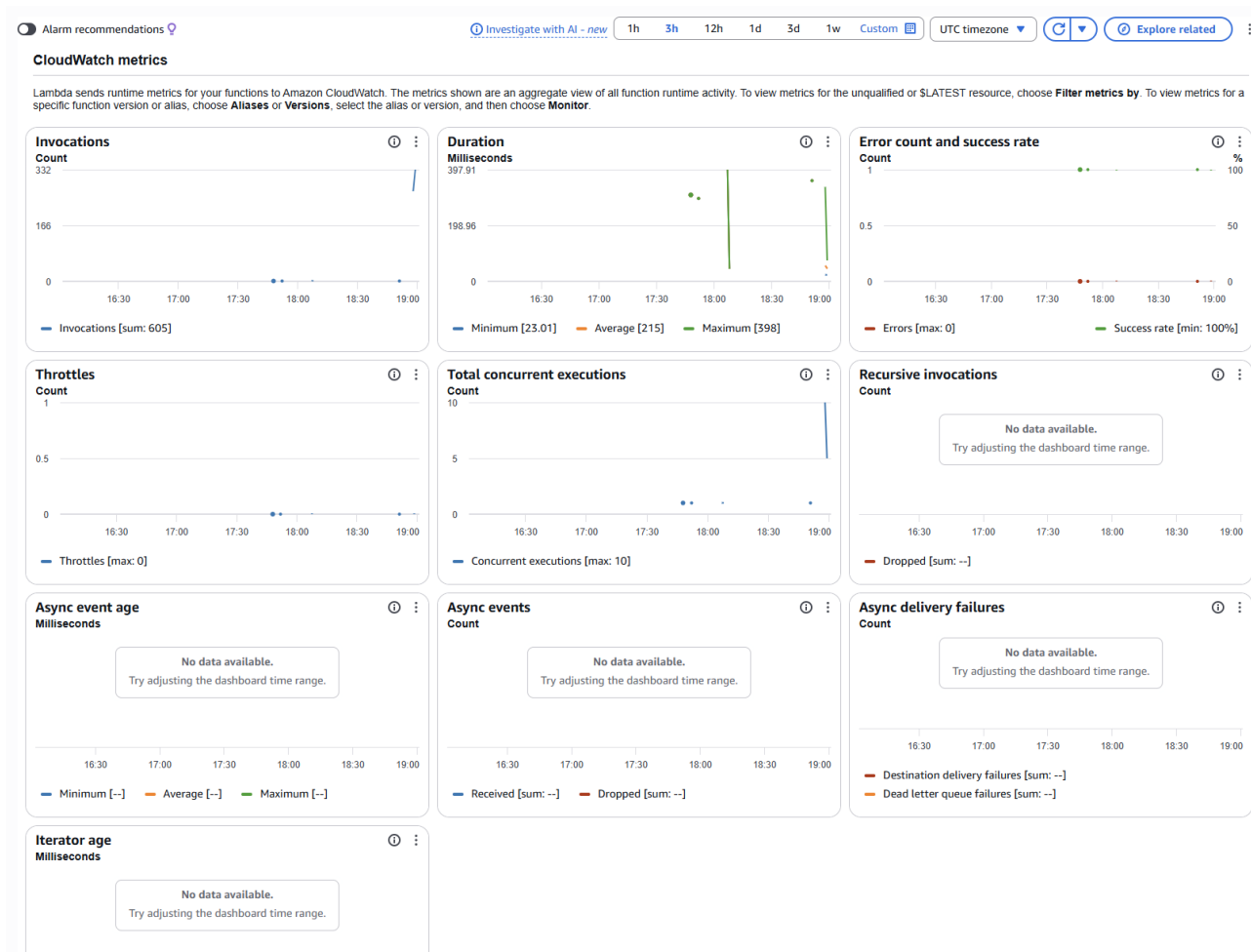
```
All VUs finished. Total time: 1 minute, 35 seconds

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Summary report @ 00:29:57(+0530)
-----

http.codes.200: ..... 600
http.downloaded_bytes: ..... 2505600
http.request_rate: ..... 5/sec
http.requests: ..... 600
http.response_time:
  min: ..... 236
  max: ..... 1396
  mean: ..... 440.2
  median: ..... 424.2
  p95: ..... 685.5
  p99: ..... 1224.4
http.response_time.2xx:
  min: ..... 236
  max: ..... 1396
  mean: ..... 440.2
  median: ..... 424.2
  p95: ..... 685.5
  p99: ..... 1224.4
http.responses: ..... 600
vusers.completed: ..... 10
vusers.created: ..... 10
vusers.created_by_name.List employee documents: ..... 10
vusers.failed: ..... 0
vusers.session_length:
  min: ..... 81531.7
  max: ..... 92590.7
  mean: ..... 87545.1
  median: ..... 86710.4
  p95: ..... 92072.4
  p99: ..... 92072.4
vusers.skipped: ..... 590
Log file: artillery-final-report.json
PS C:\EmployeeVault\LoadTests\
```

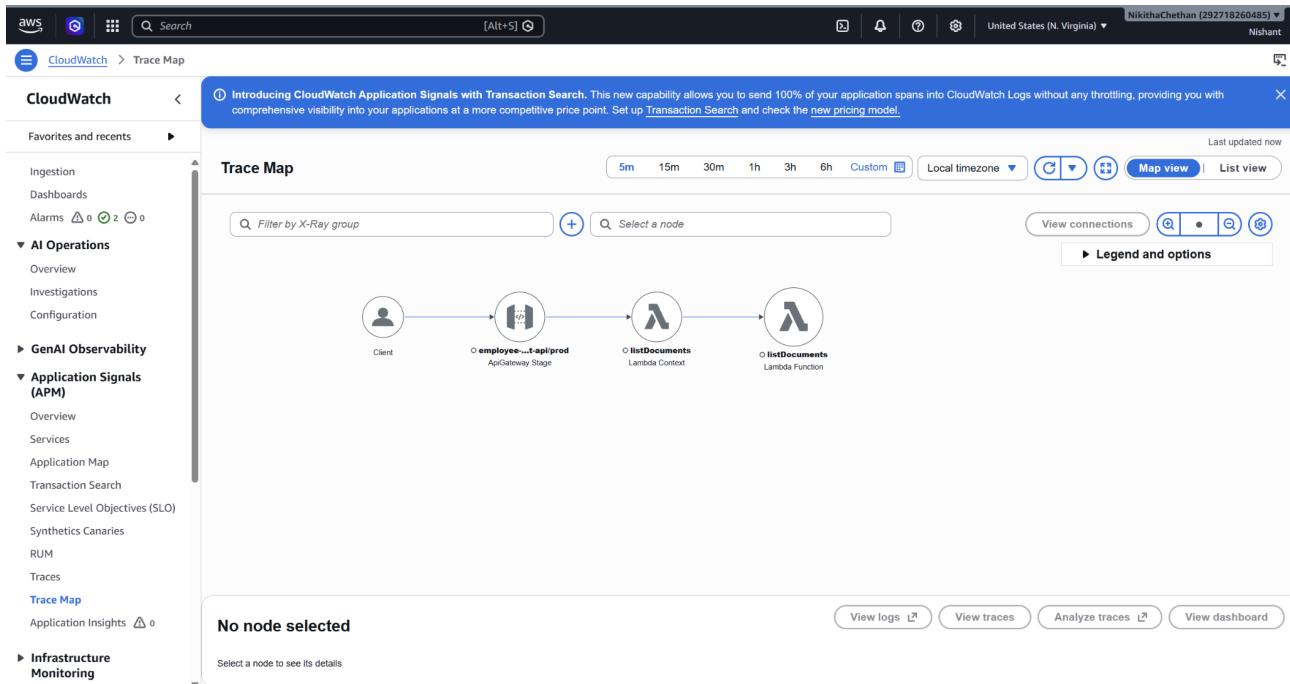
Evidence: final Artillery run showing 600 HTTP 200 responses, 0 failures, and latency percentiles.

B. CloudWatch Lambda metrics



Evidence: listDocuments Lambda metrics showing 0 errors, 0 throttles, max concurrency 10, average duration 215 ms.

C. X-Ray Trace Map



Evidence: traced request path from client through API Gateway to listDocuments Lambda.

D. GitHub Actions CI/CD success

nishnarudkar / Employee_document_vault

Code

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← Employee Document Vault CI/CD

Employee Document Vault CI/CD #2

Re-run all jobs

Latest #2

Summary

All jobs

Run Python Tests

Deploy Lambda Functions

Run details

Usage

Workflow file

Re-run triggered now

nishnarudkar -o- 486e549 main

Status

Success

Total duration

16s

Artifacts

—

deploy.yml

on: workflow_dispatch

Run Python Tests 8s

Deploy Lambda Functions 11s

Annotations

1 warning

Deploy Lambda Functions

Node.js 20 is deprecated. The following actions target Node.js 20 but are ...

Show more

Evidence: GitHub Actions run #2 on main completed successfully; both Python tests and Lambda deployment passed.

Security note: no Cognito password, bearer token, repository secret value, or other credential is included in this report.